

University of Wah

Department of Computer Science



Submit to:

Ms. Wasifa Kanwal

Submit by:

Muhammad Talha

Course:

Information & Security Lab 251L

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

Lab Task # 11 :-

Plain text:

My name is Talha.

Key:

This is a 6 Bytes Key

Plain text (Hexa):

4D, 79, 20, 4E, 61, 6D, 65, 20, 69, 73, 20

~~00, 00, 00, 00, 00, 00~~, 54, 61, 6C, 68, 61

Key (Hexa):

54, 68, 69, 73, 49, 73, 41, 31, 36,

42, 79, 74, 65, 4B, 65, 79.

Plain text matrix (column):

4D	61	69	61
79	6D	73	6C
20	65	20	68
4E	20	69	61

Key matrix (row):

w ₀	54	68	69	73
w ₁	49	73	41	31
w ₂	36	42	79	74
w ₃	65	4B	65	79

Key Expression:

Take w₃:

$$w_3 = [65 \quad 4B \quad 65 \quad 79]$$

$$\Rightarrow w_3 = [4B \quad 65 \quad 79 \quad 65]$$

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By replacing with sbox
 $w_2 = [B3 \quad 4D \quad B6 \quad 4D]$

Now taking XOR of x_3 with round constant (01, 00, 00, 00)

After doing:

$$g(w_2) = [B2 \quad 4D \quad B6 \quad 4D]$$

$$w(4) = w(0) \oplus g(w(3))$$

$$\begin{array}{cccc} \bar{54} & 68 & 69 & 73 \\ B2 & 4D & B6 & 4D \end{array}$$

$$w(4) = \underline{E6 \quad 25 \quad DF \quad 3E}$$

$$w(5) = w(4) \oplus w(1)$$

$$= [E6 \quad 25 \quad DF \quad 3E] \oplus [49 \quad 73 \quad 47 \quad 31]$$

$$w(5) = [AF \quad 56 \quad 9E \quad 0D]$$

$$w(6) = w(5) \oplus w(2)$$

$$w(6) = [99 \quad 14 \quad E7 \quad 79]$$

$$w(7) = w(6) \oplus w(3)$$

$$w(7) = [FC \quad 5F \quad 82 \quad 00]$$

Round 1: **E6 25**

99 14 E7 79 DF 3E AF 56 9E 0
 FC 5F 82 00

4D	61	69	61		EG	AF	99	EC
79	6D	73	6C	⊕	25	56	14	5F
20	65	20	68		DF	9F	E7	82
4E	20	54	61		3E	0D	79	00

NSM =

AB	CE	FO	9D
5C	3B	67	33
FF	FB	CF	EA
70	2D	2D	61

Round 1

Step 1: Sub Byte :-

Replace value by Sbox :

62	8B	8C	5E
4A	E2	85	C3
16	0F	C6	87
51	D8	D8	EF

Step 2: Shift Row

NSM =

62	8B	8C	5E
E2	85	C3	4A
C6	87	16	0F
EF	51	D8	D8

Step 3 : Fixed Matrix $\times$ NSM

02	03	01	01		62	8B	8C	5E
01	02	03	01	⊕	E2	85	C3	4A
01	01	02	03		C6	87	16	0F
03	01	01	02		EF	51	D8	D

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C3	F3	9E	41
87	A9	6F	62
1D	07	C8	2D
9E	6B	A4	7E

Add Round Keys:

C3	F3	9E	41	⊕	E6	AF	99	F1
87	A9	6F	62		2E	56	14	5F
1D	07	C8	2D		DE	9E	E7	87
9E	6B	A4	7E		3E	0D	79	00

25	5C	07	BD
A2	FE	2B	3D
C2	99	2F	AF
A0	66	DD	7E

Code

```
practice.py > ...
1  from Crypto.Cipher import AES
2  from Crypto.Random import get_random_bytes
3  import base64
4
5  key = get_random_bytes(16)
6  iv = get_random_bytes(16)
7
8  plain_text = "My name is Talha"
9
10 def pad(text):
11     while len(text) % 16 != 0:
12         text += ' '
13     return text
14
15 cipher = AES.new(key, AES.MODE_CBC, iv)
16 padded_text = pad(plain_text).encode('utf-8')
17 cipher_text = cipher.encrypt(padded_text)
18
19 cipher_text_base64 = base64.b64encode(cipher_text).decode('utf-8')
20 print("Encrypted Text:", cipher_text_base64)
21
22 cipher_decrypt = AES.new(key, AES.MODE_CBC, iv)
23 decrypted_text = cipher_decrypt.decrypt(cipher_text).decode('utf-8').strip()
24 print("Decrypted Text:", decrypted_text)
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

...

Encrypted Text: 0qP6we/rfGGv57AtduU4Dg==

Decrypted Text: My name is Talha